Biology Feedback for DnaA Replication-Initiation Report

Compact expert-review guidance for improving biological interpretation, validation targets, and mechanistic clarity.

Core message: the report has the right biological decomposition, but must separate heuristic regression from biological validation, map observables precisely, and use perturbation-based mechanism tests.

1. Scope and biological framing

This feedback reviews the biological interpretation of the DnaA / replication-initiation investigation report. The goal is to improve how the report helps experts evaluate mechanisms, not to critique the HTML implementation.

The current study decomposition is scientifically sensible: DnaA expression, autorepression, nucleotide state, chromosomal titration, oriC assembly, initiation triggering, RIDA / DDAH / DARS reset, and SeqA sequestration. The main issue is that the report sometimes treats a staged textbook narrative as more settled than the current evidence warrants.

• Separate regression compatibility from biological validation and explanatory gain.

• Map every literature value to the exact model observable before declaring pass or fail.

• Use perturbations to validate mechanism, not wild-type baseline alone.

• Make expert uncertainty explicit and actionable.

2. Priority biology TODOs

Priority	Actions
P1 -- must fix	• Classify targets as regression, biological validation, or explanatory gain. • Add literature-to-model observable mappings for every biological threshold. • Replace Pearson-only autorepression testing with a promoter-response test suite. • Define active DnaA pools explicitly: total, free, ATP-bound, ADP-bound, DNA-bound, oriC-bound. • Require perturbation panels before calling a mechanism biologically validated.
P2 -- next	• Expert-curate the DnaA-box catalog: oriC, dnaAp, datA, DARS, and generic titration sites must not be conflated. • Version SeqA abstraction levels: fixed timer, methylation-state model, cooperative SeqA-Dam model. • Clarify RIDA, DDAH, and DARS as reset mechanisms rather than generic hydrolysis knobs. • Classify biological test failures by likely cause: missing mechanism, miscalibration, observable mismatch, or insufficient statistics.
P3 -- improve	• Add mechanism cards for expert-editable assumptions. • Add "what would change my mind" boxes for major conclusions. • Turn expert references into specific answerable expert questions.

3. Compact TODO specifications

TODO	Specification	Acceptance criteria
Target class split	Every acceptance criterion declares one primary class: regression_compatibility, biological_validation, or explanatory_gain. The report renders these separately.	A heuristic-match test cannot count as biological validation. Final decisions show all three outcomes independently.
Observable mapping	Every literature comparator includes literature observable, model observable, transformation, time window, aggregation, and caveats.	No numeric literature threshold can be used without a mapping table and source condition.
DnaA abundance pools	Define whether abundance target includes free DnaA, DnaA-ATP, DnaA-ADP, apo-DnaA, DNA-bound DnaA, and complexed forms.	DnaA count pass/fail is blocked if pool mapping is unknown or expert-disputed.
Autorepression suite	Replace same-time Pearson r with lagged cross-correlation, conditional transcription probability, burst-rate by promoter state, gene-dosage correction, and dnaAp1/dnaAp2 response if available.	Sparse mRNA data can return insufficient evidence rather than false pass/fail.
DnaA-ATP decomposition	Report total, free, DNA-bound, oriC-bound, non-oriC-bound, ADP-bound, and apo-DnaA pools.	Initiation studies cannot use total DnaA-ATP alone as the active initiation signal.
Perturbation panel	Add perturbations for dnaA expression, autorepression strength, ATP cycle, titration sites, oriC affinity, RIDA, DDAH, DARS, and SeqA.	Wild-type baseline alone cannot close biological validation.
DnaA-box schema	Each site records region type, box class, affinity, nucleotide preference, cooperative group, source, and expert-review status.	oriC, dnaAp, datA, DARS, and titration boxes are distinguishable in analysis and plots.

TODO	Specification	Acceptance criteria
SeqA versions	Define SeqA_v0 fixed timer, SeqA_v1 methylation-state plus Dam, SeqA_v2 cooperative SeqA-Dam dynamics.	Claims about SeqA are limited by the implemented abstraction version.
Reset mechanism cards	RIDA, DDAH, and DARS each get biological role, model abstraction, included behaviors, omitted behaviors, perturbations, and claim scope.	Hydrolysis-rate requirements are labeled model-implied unless supported by direct biology.
Expert questions	Convert uncertainties into answerable questions with alternatives, impact if wrong, blocking status, and requested response type.	Expert input changes thresholds, mappings, or mechanism cards in a versioned way.

4. Minimal biology schemas to add

```
target_class: regression_compatibility | biological_validation | explanatory_gain

measurement_mapping:
literature_observable: {name, biological_pool, condition, statistic, source_ids}
model_observable: {listener_path, state_path, molecular_pool, units, includes_bound_species}
transformation: {formula, time_window, aggregation}
caveats: []

autorepression_tests:
- lagged_binding_mrna_cross_correlation
- conditional_transcription_probability_given_promoter_bound
- transcription_burst_rate_by_promoter_state
- dnaA_gene_dosage_corrected_expression
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5. Desired biology-report behavior

For every biological claim, the report should answer: what mechanism is being tested, what abstraction represents it, what literature observable is being compared, what model observable is measured, what uncertainty remains, what perturbation would distinguish competing explanations, and what expert decision is needed.